AURA Twin

SEE THE PATIENT. UNDERSTAND WHAT COMES NEXT.

COMPOSITE RESEARCH TWIN / aura-research-1.0.0

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A useful forecast must also be inspectable

The problem: disconnected context, changing signals and opaque confidence

01

Understand the current state

See recorded CGM, heart rate and IBI-derived variability with source and freshness.

02

Review the possible future

Four glucose horizons, an explicit event contract and measured uncertainty.

03

Challenge the calculation

Inspect the input cutoff, perturb sensors, freeze a prediction and reveal observed outcomes.

Two sources. Two identities. One honest boundary.

Composite context is not a matched clinical cohort

01

Official Synthea sample

16 adult profiles · historical cutoff 2019-01-01 · source rows and missing fields retained.

02

PhysioNet BIG IDEAs 1.1.3

16 research participants · actual CGM, HR and IBI · published-source checksums.

03

Target-blind association

Source-ID rank maps to UUID-sorted synthetic profiles. No outcome-driven matching.

Missing information stays missing

Authentic source fields, chronological filtering and visible provenance

01

Before the EHR cutoff

Only source observations and conditions available by the historical snapshot are retained.

02

Nullable clinical context

BMI absent in 1 profile, HbA1c in 9 and LDL in 6. No invented genetics.

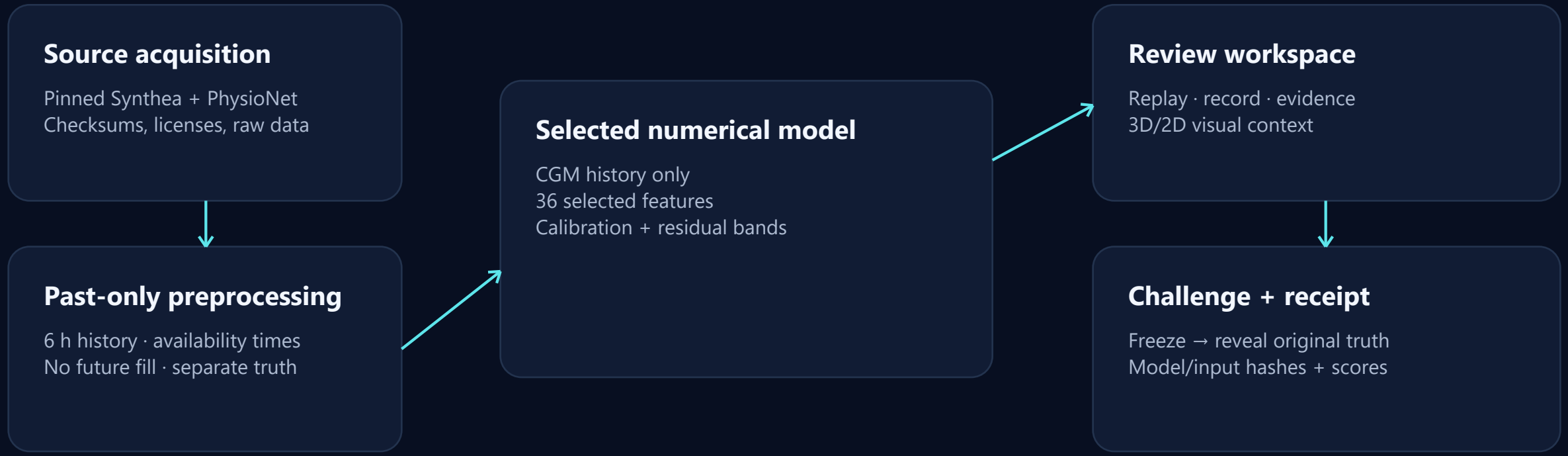
03

Unavailable sensor channels

No imported step count, activity or sleep stages. RMSSD is derived from accepted IBI pairs.

A traceable path from source to prediction

Research preprocessing and runtime inference share the same contract



State the event before showing a probability

Research event: observed CGM rise ≥20 mg/dL at approximately +120 minutes

01

Current event contract

CGM rise ≥20 mg/dL
at approximately t + 120 min

Relative to the latest available current reading.

02

Observed ground truth

Nearest real CGM sample within ±150 seconds of the nominal horizon.

No interpolated outcomes. No future input.

Separate people before evaluating windows

9 train · 2 selection · 2 calibration · 3 final participants

9

Train

Fit candidates
IDs 001–009

2

Select

Choose family/configuration
IDs 010–011

2

Calibrate

Probability, policy, bands
IDs 012–013

3

Final

Opened after freeze
IDs 014–016

The model only receives the recorded past

Six-hour history, explicit availability times and automated prefix checks

01

CGM

Backward as-of alignment; carry forward no more than 15 minutes; age and coverage are features.

02

Heart rate and IBI

Completed past bins; minute-rounded HR delayed 60 seconds; no cross-gap HRV pairs.

03

Leakage checks

Truncating future rows must preserve features. Source timestamps cannot exceed the cutoff.

Compare fusion with honest alternatives

Primary family selection excludes unmatched EHR negative controls

Family	Role	Interpretation
CGM history	Eligible	Recorded glucose history
CGM + HR + RMSSD	Eligible	Adds actual wearable channels
Matched research fusion	Eligible	Adds study HbA1c + sex
Unmatched Synthea only	Negative control	Different synthetic person
Unmatched composite fusion	Negative control	Cannot select primary family
Persistence	Comparator	Latest glucose carried forward

Measured research results, with their denominator

1,859 final windows from 3 participants

Model	+120 MAE	Recall	Precision	FN / FP
CGM only	13.31	66.20%	22.14%	97 / 668
CGM + HR + RMSSD	13.70	84.67%	19.35%	44 / 1,013
Matched research fusion	14.73	82.93%	22.88%	49 / 802
Unmatched EHR only	14.80	100.00%	15.44%	0 / 1,572
Unmatched composite	13.76	76.66%	20.52%	67 / 852
Persistence	16.35	0.00%	0.00%	287 / 0

Same eligible final windows · events are overlapping windows, not unique episodes

Confidence should be measured, not decorated

Calibration, residual bands and a frozen review policy

01

Measured coverage

93.54%

Nominal 90% empirical bands; only two calibration participants.

02

Mean interval width

71.68 mg/dL

At approximately +120 minutes. No patient-level coverage guarantee.

03

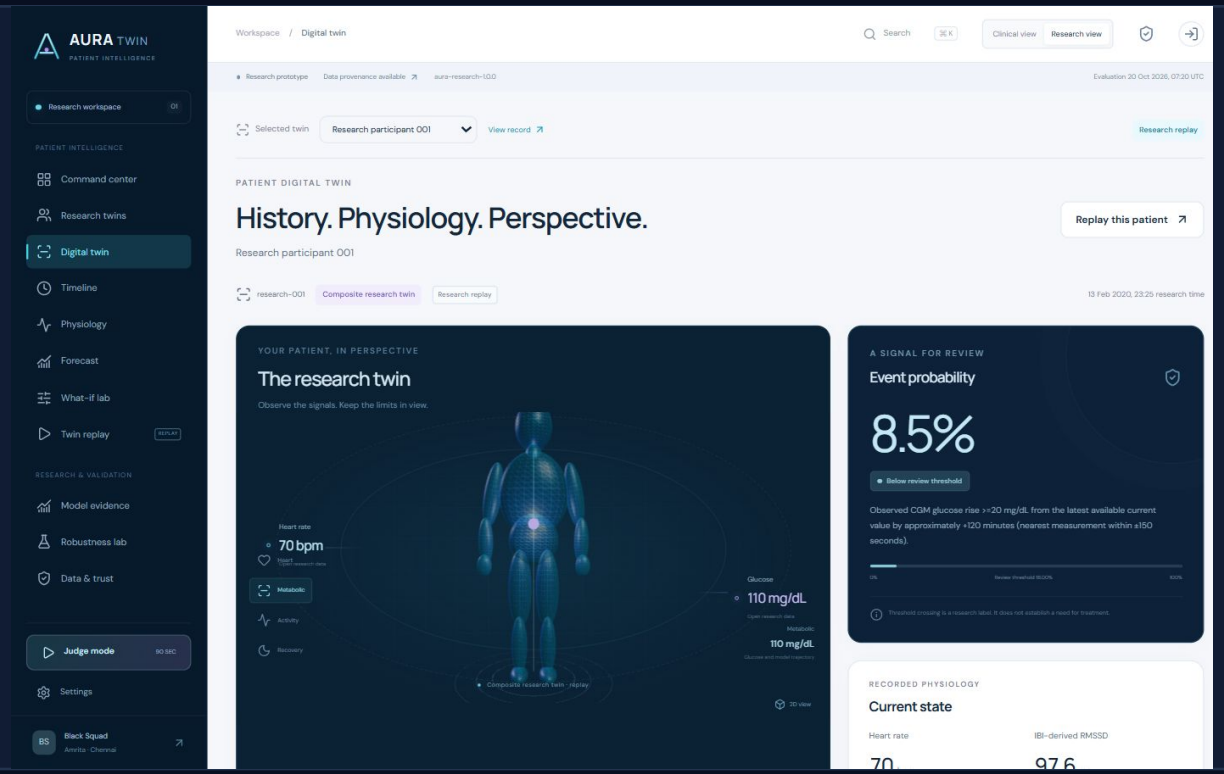
Frozen review threshold

0.180

Calibration maximum-F1 policy. Not a validated clinical trigger.

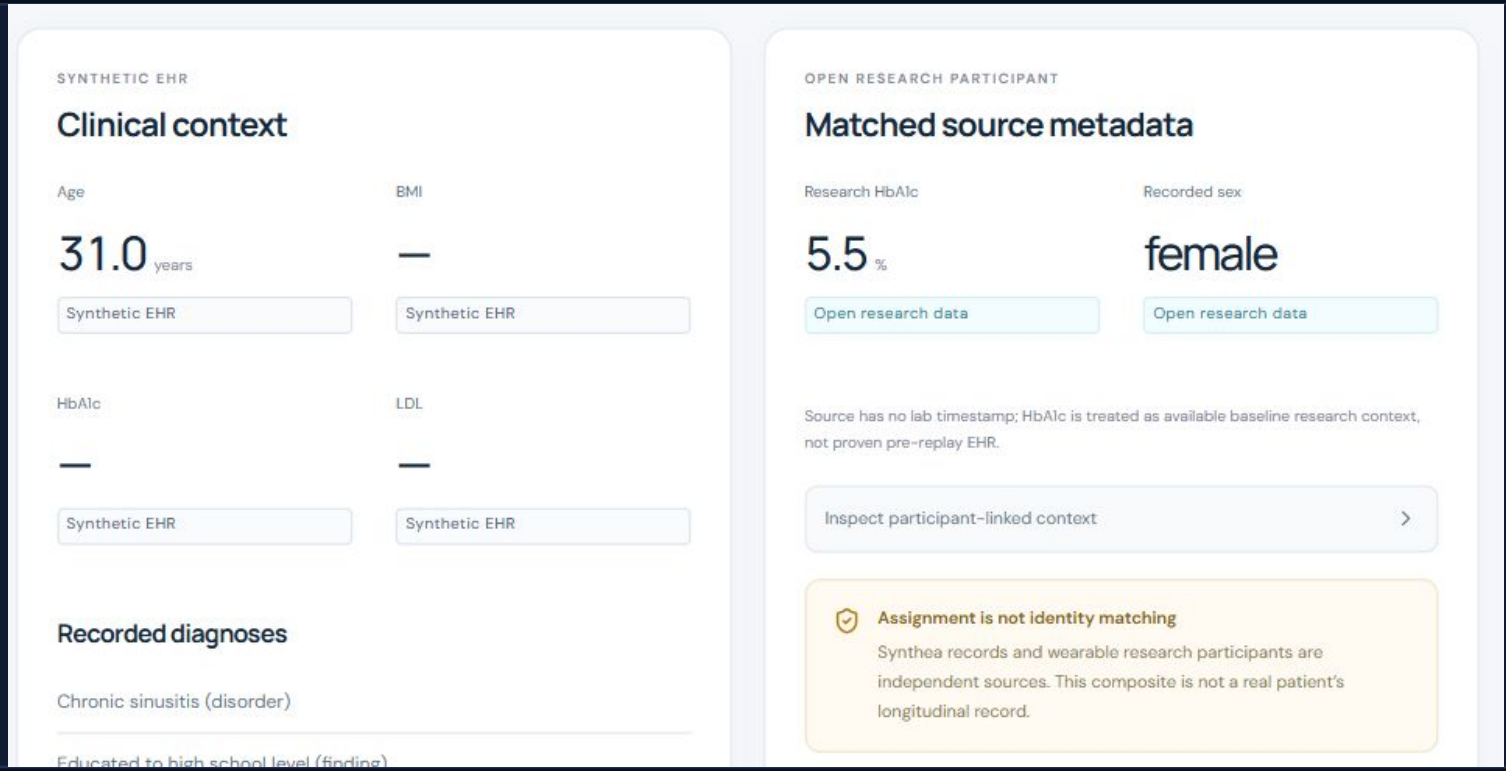
A clinician-oriented workspace

Patient context, observed signals and forecast meaning in one view



The record view keeps provenance close

Synthetic context and matched research metadata remain distinct



Explain the computation without claiming causality

Input lineage and reference sensitivity, with explicit limits

01

Show inputs used

Feature names, source channels, cutoff and availability boundary.

02

Reference sensitivity

Replace a feature group with training-reference values and recompute risk.

03

Interpretation limits

Nonadditive; not SHAP; not a treatment effect or a causal explanation.

Freeze first. Reveal later. Compare honestly.

A computation boundary the jury can inspect

01

Choose

Participant, cutoff
and input stress case

02

Freeze

Restrict the prefix
Preserve model output

03

Reveal

Original recorded future
No new prediction

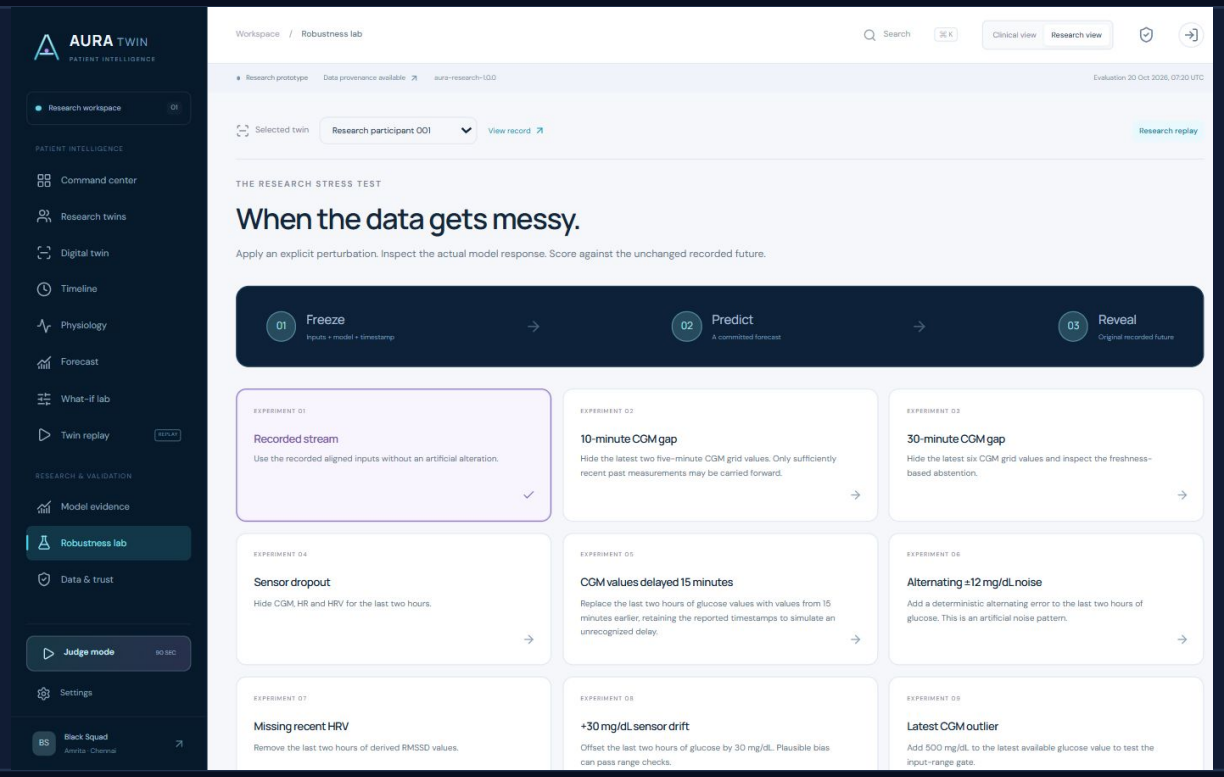
04

Compare

Error, persistence, bands
Download receipt

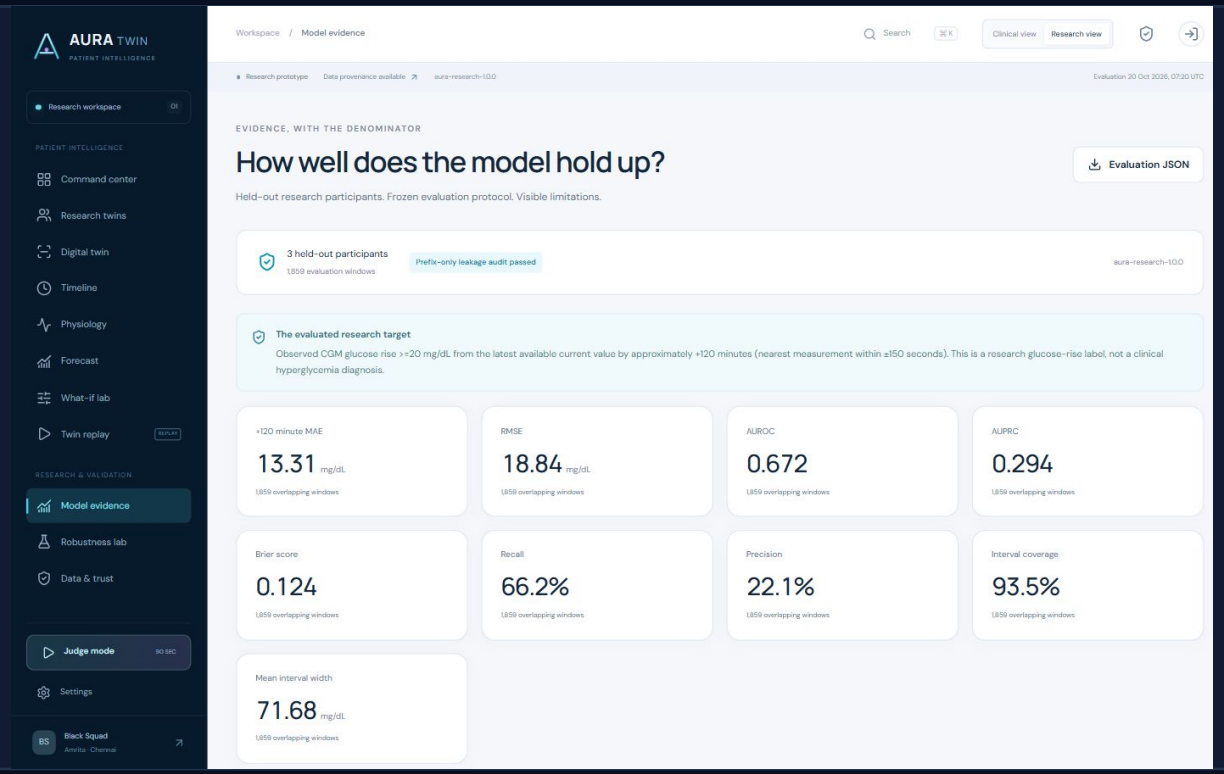
Test the failure boundary

Sensor stress is distinct from clinical robustness



Keep the evidence beside the experience

Ablations, calibration, participant slices and an explicit model contract



A working demo does not require a fake login

Demo mode, configured authentication and persistence have separate boundaries

01

Deterministic demo

Open the bundled research workspace without an account.

02

Provider-backed identity

Supabase connected. Email confirmation on; Google off. Account-flow verification pending.

03

Local and hosted state

Document which actions persist locally, remotely or only for the current session.

Separate research evidence from simulator progress

Different data and different targets cannot be compared as one accuracy claim

01

Current research model

CGM history only · open recorded wearable data · small participant cohort.

02

Legacy synthetic V3

Preserved simulator benchmark. Its prior scores are explicitly historical.

03

Independent V4 experiment

Predeclared promotion gate failed; not promoted.

Verification follows the claims

Numerical parity, chronology, source integrity and interaction checks

01

Data and model

Source checksums, patient split, prefix invariance and Python/JavaScript parity.

02

Product behavior

Replay boundary, quality states, frozen reveal, export and responsive interaction.

03

Known limits

Small cohort, unknown future events, imperfect sensors and no clinical validation.

A short, reproducible jury walkthrough

Use the interface to test the evidence

01

First 30 seconds

Open demo. Identify composite provenance, model version and replay cutoff.

02

Next 30 seconds

Inspect source inputs and forecast uncertainty. Freeze a prediction.

03

Final 30 seconds

Reveal actual outcomes, compare persistence, inspect a failure case and export evidence.

A forecast you can question.

Actual sources.
Inspectable predictions.
Honest evidence.

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